Mathematics AI HL · Set D · Paper 3

Question booklet · 75 minutes · 55 marks

Answer all questions. A calculator is permitted. Show your reasoning. Give exact answers or round to three significant figures unless directed otherwise. Write on separate paper.

1. Investigation 1 [28]

Cooling under a changing air temperature

A temperature probe starts at 34 °C. During the next 20 minutes the air temperature is $A(t) = 18 + 0.5t$ °C. The probe model is $T'(t) = -0.2(T(t) - A(t))$, with t in minutes.

(a) Find the initial rate of temperature change and interpret its sign. **[3]**

(b) Use Euler's method with a four-minute step to estimate $T(4)$ and $T(8)$. Show both updates. **[4]**

(c) For a step length h, derive a recurrence for the Euler estimates of the gap $g_n = T_n - A(nh)$. Find its fixed point. **[4]**

(d) Use the recurrence to find the range of positive step sizes for which deviations from the fixed gap converge to zero. For which of these do the deviations alternate in sign? **[4]**

(e) Verify that $T(t) = 15.5 + 0.5t + 18.5e^{-0.2t}$ satisfies both model conditions. Use it to calculate the absolute error of the four-minute-step estimate at $t = 8$. **[5]**

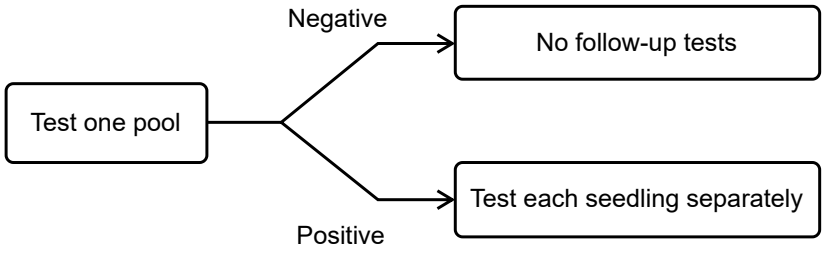
(f) Find the minimum probe temperature on the stated interval. Will the probe ever match the air temperature during that interval? Justify the connection. **[5]**

(g) A technician uses $h = 5$ and concludes that after five minutes the temperature gap is exactly −2.5 °C. Evaluate that inference using both models. **[3]**

2. Investigation 2 [27]

Pooling tests in a plant nursery

Each seedling independently carries an infection with probability 0.04. In a two-stage protocol, samples from n seedlings are pooled into one test. A positive pooled result triggers individual tests of all n seedlings. Begin with perfectly accurate tests. Use $2 \leq n \leq 10$ unless stated otherwise.



(a) For a pool of five, find the probabilities of no infections and exactly one infection. **[3]**

(b) Derive the expected number of tests per seedling as a function $f(n)$. **[4]**

(c) Use technology to find the minimizing integer pool size. Give sufficient evidence over the allowed range. **[4]**

(d) For 600 seedlings arranged in 100 such pools, find the expected total number of tests and its standard deviation. **[5]**

(e) A pooled test now has sensitivity 0.95 when a pool contains at least one infection. Its false-positive probability for a clean pool is 0.02. Individual tests are accurate. For $n = 5$, given a positive pool, find the probability that at least one seedling is infected. **[4]**

(f) Derive a revised expression for expected tests per seedling. Reassess the minimizing size. Explain why the testing cost alone is inadequate as a criterion. **[4]**

(g) A field worker puts seedlings from the same tray in each pool. Within-tray infections tend to cluster. Identify the term in the model that needs new evidence and propose a field comparison that would measure it. **[3]**

End of question booklet · 55 marks